

14 May 24

Fundamentals of Machine Learning

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Preamble: Spotify released an API that reports the audio features of its songs, such as “tempo” or “energy”. Here, we will use these features of 50k randomly picked songs to predict the genre that the song belongs to.

My task was to most accurately predict the genre of a selection of Spotify songs based the audio features of that song. First, I analyzed the dataset to get an understanding of the types of features/values I would be working with. I carefully observed the dataset and noted any missing values. I then applied the appropriate transformations on any data that were categorical or were by nature not feasible to work with in the models to come. I performed a dimensionality reduction technique, and then trained the model on the training data. Next, I initialized and trained the random forest model, and then made predictions on the test set. I obtained the AUC score which I used to assess the model accuracy. Finally, I observed the reduced dimensions via the dimensionality reduction method I performed to better understand the clusters on the data and how they contributed to the success of the model. In this paper I provide a brief outline of the analysis I performed on the Spotify dataset using the aforementioned steps. I will summarize my findings with diagrams to illustrate the impact of each of the musical features on the overall accuracy of classifying the music genre.

The initial step was to first import the necessary packages that I would be using for my analysis and to perform the preprocessing of the data.

I resolved the occurrences of missing values by replacing those with “?” with nan and by imputing those nan’s with the median values of their corresponding feature.

```
# Apply imputation for numerical columns
data.replace('?', np.nan, inplace=True)
data.dropna(inplace=True)

num_imputer = SimpleImputer(strategy='median')
data[numerical_cols] = num_imputer.fit_transform(data[numerical_cols])
```

In the next step, I encoded the categorical values of ‘mode’ and ‘key’. I also did the same for ‘acoustic’ as a means of allowing the data to be used in the classification instead of just dropping them altogether. As I will show, the features played an important role in boosting the accuracy of the model performance.

Next, I set the test set to the appropriate size specified in the spec sheet. As with traditional dimensionality reduction practice, I fit the standard scalar on the data prior to running the PCA. This was done to correct for the problem of the ‘genre’ values not being normally distributed. I applied the PCA, setting the ‘n_components’ value to .95 to properly retain a sufficient amount of the variance. I then did KMeans clustering with 5 clusters. This was done because I found that this value most prominently boosted the model accuracy in the later steps. I then split the training/test data to the proper specifications as called for in the spec sheet. I fit the random forest model with the training data, carefully ensuring that the random seed was set to my N#: 14798210 and ‘n_estimators’ to 100, as the model performed better with this number as opposed to 10 or 1000. I made predictions on the data, and evaluated the overall model performance with the AUC score. I obtained an AUC score of “0.8969154728910118” with the PCA.

To visualize the clusters of the data in the lower dimensionality space, I plotted the PCA with 2 components, as most of the model variance was explained in this space and I found that

model performance actually decreased when increasing the number of components in the PCA analysis.

In the plot, I found that the colors mostly correlated to 2 separate and distinct colors, with a bit of overlay between other separate clusters. This implies that the PCA effectively captured the variance of the data that correlates to the respective categorization. Overall, the PCA did an effective job in explaining the variance in the lower dimensional space.

Here is the ROC curve for my analysis:

